

GEMINATION AND ALLOMORPHY IN THE PROTO-
GERMANIC MN-STEMS: *BOTTOM* AND *RIME*

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§1 In the Germanic languages there is limited but significant evidence for the occasional loss of the labial nasal in the Indo-European suffix *-men-*. This development is reminiscent of the situation in Indo-Iranian. In *Die Nominalsuffixe* (*Altindische Grammatik* II, 2), Wackernagel and Debrunner mention an interchange between *mn-* and *n-* stems in Sanskrit. According to them, this interchange reflects a Proto-Indo-European rule of *mn* becoming *n*: “So gehören die Instrumentale v. *prathinā prenā bhūnā mahinā* [...] zu *prathimán-premán- bhūmán- mahimán-* [...] und beruhen auf grundsprachlichem Wandel von *mn* zu *n* hinter labialem Anlaut [...]” (§ 609a.).

The evidence for the rule is not limited to Sanskrit. There are also cases from other languages that show loss of an *m* in original stems in *-men* or *-mon*. In all of these cases, the root contains a labial:

- PIE **b^hud^h-nó-* ‘bottom’: Lat. *fundus* m., OI *bonn* m., Skt. *budhná-* m. ‘bottom’. In view of Gr. *πυθμήν* m. ‘bottom’, the original suffix may have been **-mno-* in the genitive (Lühr 2000: 301-302).
- Av. *raoyna-* ‘butter’. If it is related to Icel. *rjómi* m. ‘cream’ < **reu(g)man-* and MHG *rōme* m. ‘id.’ < **rau(g)man-*, *raoyna-* may derive from **(H)re/ou^h-mno-*. Pokorny (p. 873) reconstructs **reug^hm(e)n-*.
- PSlav. **věno* n. ‘dowry’ (= SCr. *vijèno* n., Russ. *véno* n., cf. Hamp 1970), Gr. *ἔδνα, ἑδνα* n. pl. ‘dowry’, OFri. *wettma* m. ‘dowry’. The root seems to be **h₁ued-*. Frisk (1, 441-442) considers the possibility that the Greek and Slavic suffixes reflect **-mno-* in view of the Germanic *mn*-suffix.
- PGm **bragna-* ‘brain’. On the basis of the Germanic forms and Gr. *βρεχμός* m. ‘skull’, Hom. *βρέχμα* n. ‘id.’, Pokorny (p. 750) reconstructs **mreg^h-(m)no-*.

The examples form a significant enough basis to assume that after roots with a labial, the cluster **-Cmn-* was simplified to **-Cn-*. The extant forms with retained *m* must consequently be due to the nomi

native, or other forms with full grade of the suffix. Especially in view of **b^hud^h-nó-*, we can date back the cluster simplification to the proto-language with some confidence.

The age of the dissimilation rule is nonetheless irrelevant to what is at stake here: it appears that the loss of *m* can account for the problematic consonantal interchange *m* ~ *p* in the word for *rime*, and the *t* or *tt* in the word for *bottom* in Germanic. In these lexemes, the loss of *m* gave rise to a combination of a stop and following dental nasal, which ultimately resulted in an allomorphic geminate in Proto-Germanic.

§ 2 Proto-Germanic geminates were first referred to by Hermann Paul (1880: 133 fn.), and explained by Ferdinand Kluge in 1884. The process of Proto-Germanic gemination has remained a fundamental issue in Germanic and Indo-European linguistics ever since. Not only does the rise of geminates constitute an essential phonological demarcation line between Proto-Indo-European and Proto-Germanic, as PIE did not have long consonants, but it is also vital to any interpretation of the two other basic sound shifts, Grimm's and Verner's law, i.e. the lenition and voicing of PIE plain stops after unaccented syllables (cf. Kortlandt 1991).

In 'Die germanische Consonantendehnung' (1884), Kluge analyzed the Proto-Germanic geminates from an Indo-European perspective, and his analysis has now become the canonical view. Even though the bulk of the words containing a geminate do not have cognates outside Germanic, Kluge, as a true Neogrammarian, concluded that the Germanic long stops had arisen through regular sound change: many Proto-Germanic geminates can be associated with *n*-suffixes in other Indo-European languages, or otherwise with an attested or putative *n*-stem paradigm in Germanic. Kluge therefore proposed that all PIE stops merged into one and the same voiceless geminate when followed by an *n*-suffix.

- OE *bucca* m. 'bock' < **bukkan-* < PIE **b^huǵon* ~ **b^huǵnós* to Arm. *buc* 'lamb', Av. *būza-* 'goat'.
- OE *þaccian* 'to hit, touch' < **pakkōjan-* < PIE **th₂g-n-* to Lat. *tangō* 'to touch', Gr. τεταγών 'taking'.
- OE *liccian* 'to lick' < **likkōjan-* < PIE **liǵ^h-n-* to Lat. *lingō*, Gr. λιχνεύω 'lick'.

- ON *hōttr* ‘hat’ < **hattu-* next to **hōda-* ‘hood, hat’. The sequence *-CCu- points to an acc. pl. *-nns of an unattested *n*-stem (cf. Lühr 1988: 200).
- MoG *schnitzen* ‘to carve’ < **snitt-* < Pre-PGm. **snit-n-* alongside *schneiden* ‘cut’ < **snīpan-*.

Parallel to the assimilation in clusters with resonants (Sievers 1878: 149 fn), Kluge concluded that the *n*-suffix had to be stressed because in some cases without a geminate an Indo-European *n* is preserved. These forms often have voiceless fricatives, which requires that they were root-stressed before Verner’s law (Kluge 1884: 169). An important instance of a PIE *n* being retained is PGm. **swefnaz* ‘sleep’. Since it only has root-stressed cognates (Skt. *svápna-*, Lith. *sāpnas* (2)¹, Gr. ὕπνος, Alb. *gjumē*²) it can be brought forward to corroborate oxytony as a necessary condition to Kluge’s law.

§ 3 Some time after Grimm’s law, long stops were shortened in heavy syllables (Paul 1880: 133; Kluge 1884: 183). Such a chronology can be deduced from the fact that geminates were not lenited by this law neither in light nor in heavy syllables. There are numerous examples of shortening³:

- ON *hvítr* ‘white’ < °*hwītta-* < PIE **kueit-no-* to Skt. *śvítna-*, *śvetá-* ‘white’ (Kluge 1884: 183).
- Go. *þairko* f. ‘hole in a needle’ < °*þerkkōn-* to Go. *þairh* ‘through’ (Lühr 1988: 269).
- OHG *scāf* n. ‘sheep’ < °*skēppa-* to Go. *skaban skof skabans* ‘to shave’ (Woods 1919: 207; cf. Davis 1991).

§ 4 On the morphological level, Kluge’s law had far-reaching consequences. There must have been a period of time during which at least some *n*-stems had two allomorphs: the nominative stem without and oblique cases with a geminate. An impressive discussion of the material is given by Lühr (1988). In short, one could say that some

¹ Following Illič-Švityč 1979: § 9. Standard Modern Lithuanian has a.p. 4.

² On initial PIE **s* > Alb. *gj* in stressed syllables, see Kortlandt 1987: 219–226.

³ In order to facilitate the discussion of the material, I shall often give the intermediate stage directly preceding heavy syllable shortening. This is also regularly done by Lühr in her *Expressivität und Lautgesetz im Germanischen* (1988). For the sake of clarity, I shall mark forms that reflect this stage with a degree mark °.

time in late Proto-Germanic, there was a paradigmatic interchange of long and short stops. The individual daughter languages have often generalized only one allomorph:

I		II		III
* <i>gnútōn</i>	>	* <i>knubōn</i>	→	OHG <i>knodo</i> m. 'knot'
* <i>gnutnás</i>	>	* <i>knuttaz</i>	→	OE <i>cnotta</i> m. 'knot'

§ 5 The generalizations of the kind observed here are well-understood. A different sort of analogy was proposed by Lühr (1988: 330-344) in her chapter on counter examples of *n*-gemination, e.g. Go. *unairkns* 'true', ON *feikn* 'terrible', Go. *taikns* 'token', etc. While these cases seem to have escaped the effects of Kluge's law, they require an explanation. Lühr assumes that the words under discussion are the outcome of no longer attested *n*-inflections, which combined the oblique geminate with the *n* of the nominative in one root form:

I		Ila		Ilb
* <i>daikōn</i>	>	° <i>taih/gōn</i>	→	PGm. * <i>taikna-</i>
* <i>daiknás</i>	>	° <i>taikkaz</i>		

Since, nevertheless, **taikna-* is not directly attested as an *n*-stem, and it cannot be explained why the adjectival weak inflection led to double analogy forms only in **erkna-* and **faikna-*, Lühr's view is difficult to maintain. In the case of **taikna-*, it seems more probable to me that the root °*taikk-* is the result of *n*-gemination in the verb: OE *tēcan* 'to teach' < °*taikkjan-* < PIE **doik-n-* (cf. Go. *gateihan* 'to announce', Gr. δείκνυμι 'to show', Av. *daēsa-* m. 'sign'). PGm. **taikna-* 'sign' is therefore best interpreted as a more recent derivative of this verb with a *na*-suffix. Apparently, this suffix was still productive after the operation of Kluge's law. Many of Lühr's other counter examples are probably the result of this derivational process.

§ 6 It must be stressed that not all long stops can be explained by Kluge's law. Geminates that occur in words with no Indo-European etymology are far from rare, e.g. OE *rætt* m. 'rat', OFri. *rokk* m. 'skirt', Go. *skatts* m. 'money'. The Indologist and Indo-Europeanist F. Kuiper in his article 'Gothic *bagms* and Old Norse *ylgr*' (1995) called "consonant variation" the most characteristic substratum feature in the Germanic lexical material. Similarly, geminates are frequently found

in Celtic words of unknown origin. A well-known example of this is the Celtic word for son, which in Ogham Irish has a genitive *maqqi*, and turns up in OIr. as *macc* (Vendryes: M1-2).

Another word with a geminate is OIr. *brocc*, MW *broch*, Bret. *broc'h* 'badger'. This word is reconstructed for Proto-Celtic as **brokko-* by Peter Schrijver (1995: 460), and has no Indo-European etymology. A link outside Celtic is nevertheless provided by OHG *braccho* 'sniffer dog' and cognates (Van Wijk 1971: 89a).

The badger (*Meles meles*) is known to have a strong sense of smell, of which it makes use to find insects while rooting in the ground. This meaning is compatible with the one found in Germanic: "sniffer dog". The formal comparison of PGm. **brakkōn* and PCelt. **brokkos* is unproblematic too. With no good Germanic and Celtic etymology present, it is therefore likely that **brokk-* was borrowed into these languages from an unknown substratum language with long stops.

It is interesting to see that **brokk-* was assigned to the class of the *n*-stems in Germanic. Geminates were apparently associated with the *n*-stems at the time when the word entered the language. Hence we may conclude that Kluge's law was anterior to the adoption of **brokk-* from an external source.

§ 7 I shall now discuss the Germanic material that evidences root variation in the words for *bottom* and *rime*. This root variation, or allomorphy, was caused by loss of *m* in the oblique cases and the operation of Kluge's law.

a. PGm. **buttma-* 'bottom'

This Germanic word shows a number of root variants:

- *bud-*: - OE *bodan* m., ME *bodme*, *boden*
 - OFri. *bodem*, *boden* m.
 - OS *bodme* dat. sg. m.
 - MLG *bōdem(e)*, *boddem(e)*, *bodden* m.
 - MDu. *bodem*, *boden*, *boom* m., Du. *bodem* c.
- *but(t)-*: - OE *botem* m., ME *botme*, *bot(t)em*, E *bottom*
 - ON *botn* m.
- *buþ-*: - OHG *bodam* m., MHG *bodem*, *boden* m., G *Boden* m.
 - OS *bothme* dat. sg. m.
 - ME *bothme*

A sound Indo-European etymology is provided by Gr. *πυθμήν* m. ‘ground, basis’, Skt. *budhná-* m. ‘bottom, ground, depth’, Lat. *fundus* m. ‘ground, bottom’, and Mlr. *bonn* ‘bottom’ (see § 1).

With regard to the morphology of the word, there are several indications that it originally had a suffix *-men-* (cf. Pokorny, 174). First of all, the Greek word attests this suffix. Secondly, in Germanic, the parallelism of the formations that continue **ēpma(n)-* ‘breath’, e.g. OFri. *ēthma* m. and OE *æðom* m. suggests such a conjugation. For these reasons I reconstruct **budman-* for Germanic as well. MLG *bod(d)eme* (Gallée 1903: 35) may be a remnant of this weak noun.

The PIE root of bottom was **b^hud^h-*, with two mediae aspiratae. The Germanic forms with **t* and **þ* are therefore unexpected. As a solution, Kluge suggested that OE *botm* is a contamination of **budman-* with a form **butta-* = Skt. *budhná-* (1884: 172 fn. 1).

Needless to say that it is by all means implausible that the geminate of **butma-* is to be explained from gemination before *m*. If this were the case, many more geminates would have been expected in this position, yet we find OHG *fadam* ‘embrace’ < **ph₂t-mo-*, OHG *rotmo* ‘blush’, ON *roðmi* ‘id.’ < **PIE *h₁rud^h-mon-*. The different effects of *m* and *n* point to assimilation of the latter during Kluge’s law, and retention of the labial nasal.

I therefore think that Kluge’s insight is correct. The fact that **but(t)ma-* received its *t* analogically from **buttaz* can nevertheless only be understood if the two root forms were still part of one and the same paradigm after Kluge’s law. In other words, the roots **bud-* and **but(t)-* must have been two allomorphs at a certain stage. In this way, Kluge’s proposal fits into the picture of a PGm. inflection in *-mn-* too. Just as the thematic formation **b^hud^h-nó-* in Sanskrit and Italo-Celtic, the Germanic allomorph **butt-* must be explained from loss of the *m* in the genitive and some other oblique cases: **b^hud^h-mnós* > **b^hud^h-nós* (cf. Lühr 2000: 301–302), which by Kluge’s law became **buttaz*.

I		II		III
<i>*b^hud^hmē/ōn</i>	>	<i>°budmē/ōn</i>	→	MDu. <i>bodem</i> , OE <i>bodan</i>
<i>*b^hud^hnás</i>	>	<i>°buttaz</i>	→	OE <i>botem</i> , ON <i>botn</i>

The paradigm remained intact until after the breaking up of Proto-Germanic. The separate daughter languages show several ways of resolving the allomorphy.

The geminate that arose by Kluge's law spread to the nominative allomorph in Old English *botem*, and probably also in Old Norse *botn*. The original dental was retained in OE *bodan*, OFri. *bodem*, *boden*, MDu. *bodem*, *boden*, *boom* m., and therefore probably also in MLG *bod(d)em(e)*. If the root had been **buþma-*, one would have expected to find forms with an *s* in (dialectal) Dutch; cf. *kossem* 'lobe, dewlab' < **kuþma-* to G *Koder* 'double chin, crop' (van Loey 1964: § 50).

Finally, OS *bothme* dat. sg. m., OHG *bodam* m. (compare also **weþman-*: OHG *wedamo* m. 'dowry') have **þ* which is problematic, but must be secondary by any account. It has been explained by Kluge (1883) as the result of a West-Germanic shift from **ð* to **þ* before an *m*, whereas Lühr (1988: 341) considered it analogical. Yet, whatever its origin may be, it has no consequences for the *mn*-paradigm reconstructed here.

The suffix also displays some variation, i.e. between *m* and *n*. Final *m* was apparently assimilated to *n* in many languages (Fick 1909: 275): OE *bodan*, OFri. *boden* (compare OFri. *gathen* 'room' < **gaþma-*), MDu. *boden*, MHG *boden*, G *Boden*. ON *botn* is probably due to assimilation too. Alternatively, it can be analyzed as a typical Scandinavian thematization of a secondary *n*-stem nom. **budmōn* ~ gen. **buttnaz* like in *nafn* n. 'name' and *vatn* n. 'water', two words that represent an Indo-European *men*-stem and a heteroclitic, respectively. There is therefore no direct evidence in favor of a reconstruction **budna-* in Germanic (pace Lühr 1988: 340-341).

b. PGm. **hrīma(n)-* ~ **hrīpan-* 'hoar-frost, rime'.

Another interesting case is the word group of **hrīma(n)-*: OE *hrīm* m., ON *hrím* n., *hrími* m., and **hrīpan-*: OS *hrīpo* m., OHG *hrīffo* m.

We may interpret the attested variation as resulting from a split paradigm **krīPmōn* ~ **krīP(m)nás*. The alternation of *m* with *p* does not indicate substratum influence, but is reflecting this old paradigm. While in the nominative **hrīPmōn* the labial stop or fricative assimilated to the *m* (cf. **hrīp-m-*, Orel 2003: 187b), the genitive form **krīPmnos* developed into **krīPnás* anterior to the operation of Kluge's law. As a result of this law, **krīPnás* became *°hrīppas*, and a new root variant arose.

The fact that the genitive form, which after the operation of Kluge's law had lost its *n* of the suffix, was reassigned to the class of the *n*-stems (= **hrīpan*) shows that the old *mn*-paradigm remained intact as an allomorphic *n*-stem during a certain period of time. The genitive

**hrīpaz* was still perceived as part of the paradigm, and accordingly acquired a new nominative **hrīpōn* in the German dialect continuum. At this point, the paradigm split into two lexemes with two different roots. The formal divergence in the daughter languages suggests that this happened after the breaking up of Proto-Germanic.

I		II		III
<i>*krīPmōn</i>	>	<i>°hrīPmōn</i>	→	ON <i>hrīmi</i> , OE <i>hrīm</i>
<i>*krīPnás</i>	>	<i>°hrīppaz</i>	→	OS <i>rīpo</i> , OHG <i>hrīffo</i>

The origin of PGm. **hrīPman-* is unknown. Even though it looks like an old formation, no cognates outside Germanic are attested. If the word were of Indo-European origin, one would perhaps expect **kreip-mon-*. At least the PGm. pair **hrīp-* ~ **hrīm-* cannot be used anymore to support Pokorny's root etymology **krei-b-*, **krei-m-*, akin to Latv. *krēims* m. 'cream', etc. (Pokorny, 618).

§ 8 Summarizing, we can state that Germanic **budman-* and **hrīPman-* have developed allomorphic paradigms because of the loss of the labial nasal and the operation of Kluge's law. According to Wackernagel and Debrunner, the loss of *m* in the clusters can be ascribed to an Indo-European rule confined to roots with an initial labial. In view of Av. *raoyna-* and PGm. **hrīPman-*, it can be claimed that the place of the labial is irrelevant to the operation of the rule.

Kluge's law was operative in the genitive and other oblique cases of stems in *-man-* where *m* was lost. This resulted in allomorphs with a geminate. Due to this new allomorphy, the paradigms of 'bottom' and 'rime' split into two or more lexemes that could compete with each other until well after the Proto-Germanic period. This adequately accounts for the dialectal consonantal variation of *m* and *p* in 'rime' and of *d* and *tt* in 'bottom'.

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